

Lecture25_
MA5104

MA 5104: Statistical Inference and Learning

Module 4: Learning

Lecture 25: Estimating f — Bias, Variance, and DimensionDr. Rakhi Singh
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- Train
Test splitsof every $\mathcal{D}$ - Build models on train, test them
on test set for selecting better
model

Two Geometries

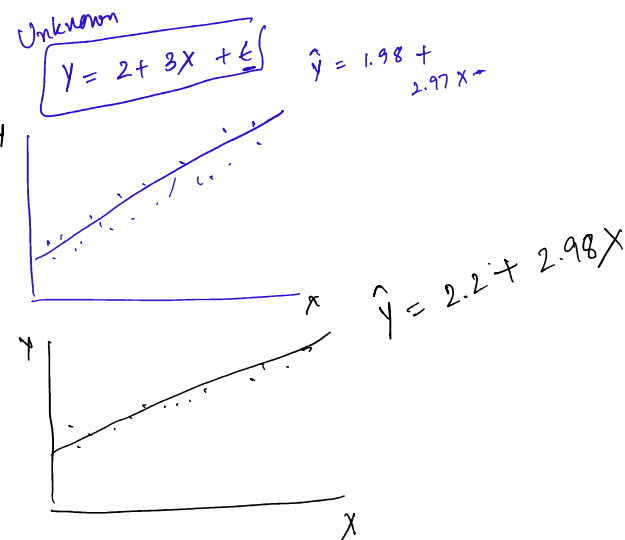
A Geometric View of Learning

Learning is not just algebra; it is geometry in the space of inputs.

- Every method draws a **shape** through the data cloud: a p -dimensional space for p features, and a way of laying the response on top of it.
- Some methods look at the whole space at once; others only at what is nearby. **Global versus local structure.**
- Today we contrast the two extremes:
 - a rigid global shape — **least squares**;
 - a flexible local shape — **k nearest neighbours**.

Same data. Very different geometries.

2/15



Least Squares: One Shape Everywhere

$$\hat{y}(x) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$$

- ▶ The model picks the best single global direction through the cloud — a **projection** onto a linear subspace.
- ▶ **Every** observation influences the fit everywhere. Move one point and the whole surface tilts slightly.
- ▶ Works when the truth is simple and smooth; it cannot bend to follow curvature. That inability is **bias**.

A single linear function is used at every point of the input space, whether or not the data there justify it.

3 / 15

kNN: Let the Neighbourhood Decide

$$\hat{y}(x) = \frac{1}{k} \sum_{i \in N_k(x)} y_i$$

- ▶ The prediction at x depends only on the responses of its k nearest neighbours. **Many small rules instead of one big one.**
- ▶ Different regions get genuinely different predictions: piecewise structure, adaptivity.
- ▶ Small k follows the data closely; large k smooths. **One knob, and it is the only one.**

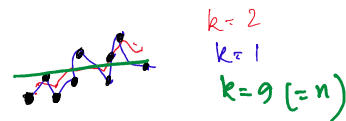
Notice what this is

Our target is $\mathbb{E}[Y | X = x]$ — the average of Y near x . k NN averages the y_i near x . It is the most literal possible attempt at the quantity we derived last week.

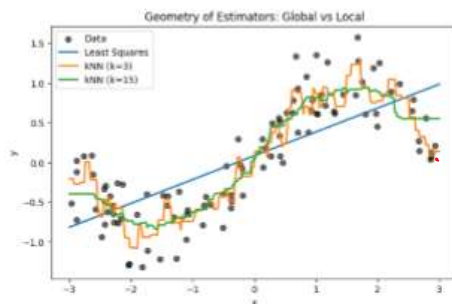
4 / 15

one hyperparameter

how do you decide its value?



Global versus Local, on the Same Data



Least squares (one line); k NN with $k = 3$ (jagged, follows everything); k NN with $k = 15$ (smoother, but flattening at the edges).

Which is right? Nothing on this plot answers that. The question is which one will do better on data we have not seen.

5 / 15

$$MSE = \|y - \hat{y}\|_2$$

$$Bias = \|\mathbb{E}(\hat{y}) - y\|_2$$

$$Var = \|\hat{y} - \mathbb{E}(\hat{y})\|_2$$

Geometry Already Explains Bias and Variance

Rigid surface (least squares)

- ▶ one global shape everywhere
- ▶ ignores local curvature $\Rightarrow$ **high bias**
- ▶ barely moves if you resample the data $\Rightarrow$ **low variance**

Flexible surface (k NN, small k)

- ▶ bends to follow nearby points
- ▶ captures local structure $\Rightarrow$ **low bias**
- ▶ changes a lot if you resample $\Rightarrow$ **high variance**

The trade, informally

Learning is choosing how much freedom the surface should have. Too stiff and it misses structure; too flexible and it follows noise.

Now let us make this an identity rather than a picture.

6 / 15

The Bias-Variance Decomposition

The Decomposition

Fix a point x_0 . Let $Y = f(x_0) + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$, $\text{Var}(\varepsilon) = \sigma^2$, and let $\hat{f}$ be fitted on a random training set $\mathcal{D}$.

$$\mathbb{E}_{\mathcal{D}, \varepsilon} \left[(Y - \hat{f}(x_0))^2 \right]$$

Insert $\mathbb{E}_{\mathcal{D}}[\hat{f}(x_0)]$ and expand exactly as in Lecture 1:

$$= \underbrace{\sigma^2}_{\text{irreducible}} + \underbrace{\left(\mathbb{E}_{\mathcal{D}}[\hat{f}(x_0)] - f(x_0) \right)^2}_{\text{bias}^2} + \underbrace{\text{Var}_{\mathcal{D}}(\hat{f}(x_0))}_{\text{variance}}$$

All three terms are non-negative, and the first is untouchable. Reducing one of the other two generally **raises** the other — this is an identity, so flexibility is always a trade, never a free improvement.

8 / 15

The Bias-Variance Decomposition

Instantiated: k NN Has Both Terms in Closed Form

With the k nearest neighbours $x_{(1)}, \dots, x_{(k)}$ of x_0 treated as fixed,

$$\mathbb{E}_{\mathcal{D}}[\hat{f}(x_0)] = \frac{1}{k} \sum_{\ell=1}^k f(x_{(\ell)}), \quad \text{Var}_{\mathcal{D}}(\hat{f}(x_0)) = \frac{\sigma^2}{k}$$

and therefore

$$\text{MSE}(x_0) = \sigma^2 + \left[f(x_0) - \frac{1}{k} \sum_{\ell=1}^k f(x_{(\ell)}) \right]^2 + \frac{\sigma^2}{k}$$

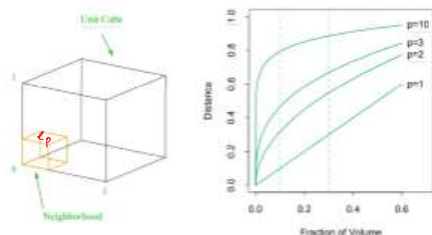
- ▶ k small: variance near σ^2 , bias near zero.
- ▶ k large: variance falls like $1/k$; bias grows, because distant neighbours have different f values.

One knob moves both terms, in opposite directions. Remember what the bias term depends on: *how far away the neighbours are*.

9 / 15

Local Neighbourhoods Stop Being Local

Let $X \sim \text{Uniform}$ on the p -dimensional unit cube. To capture a fraction r of the data, a hypercube neighbourhood needs edge length $e_p(r) = r^{1/p}$.



At $p = 10$, capturing just 10% of the data requires $e_{10}(0.1) \approx 0.80$: **eighty percent of the range of every single coordinate**.

A neighbourhood that was local in low dimension has become global.

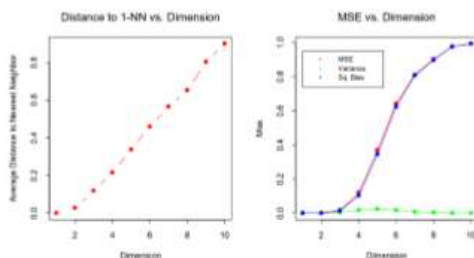
11/15

The Consequence for the Bias Term

Recall

$$\text{Bias}(x_0) = f(x_0) - \frac{1}{k} \sum_{\ell=1}^k f(x_{(\ell)})$$

This is small only if $f(x_{(\ell)}) \approx f(x_0)$ — that is, only if the neighbours really are *near*.



12/15

Why Linear Models Escape

Suppose the truth really is linear, $Y = X^T \beta + \varepsilon$. For least squares, the expected prediction error

$$\mathbb{E}[\text{EPE}] \approx \sigma^2 \left(1 + \frac{p}{N}\right)$$

- **No bias**, because the model class contains the truth.
- Variance grows only **linearly** in p — not exponentially.
- The strong structural assumption is what prevents the blow-up.
- **But the assumption must hold**. If it does not, there is no guarantee at all.

Structure replaces locality as the defence against dimension.

13/15

So What Do We Do Instead?

High-dimensional problems are solvable only because real data are not arbitrary. Instead of estimating an arbitrary $f(x_1, \dots, x_p)$, we assume

$$f \in \mathcal{F}_{\text{restricted}} \subset \mathcal{F}_{\text{all}}$$

- ▶ **Smoothness**: nearby inputs give nearby outputs.
- ▶ **Sparsity**: only a few coordinates matter.
- ▶ **Additivity**: $f(x) = \sum_j f_j(x_j)$ — no high-order interactions.
- ▶ **Low-dimensional structure**: the data lie near a manifold or a subspace.

Smaller classes reduce variance; the price is bias if the restriction is wrong.
Every remaining lecture is a different bet about which restriction is true.